Screening the *Formica* hybrid zone for intrinsic incompatibilities: a structure-corrected two-locus association scan

(work in progress)

Work-in-progress note. This document records methods, reasoning and interim results for the intrinsic (Dobzhansky–Muller) arm of Module D, beyond the descriptive among-population summary reported separately. Everything here is structure-corrected but pre-simulation: a neutral, recombination-matched null (Module E) is still required to convert the leads below into excess-over-neutral statements. Numbers may change as that null and the analyses mature.

1 What an intrinsic incompatibility looks like in replicate hybrids

The design is twenty independent, non-migrating hybrid populations, each sorting *F. aquilonia*/*F. polyctena* ancestry out of the same admixture. Against that background an intrinsic incompatibility can leave three distinct traces, and separating them is what organises the analysis.

(1) Unidirectional sorting. An incompatibility in which one parental combination is fitter drives the *same* parent's ancestry toward fixation repeatedly across replicates. This is the per-locus parallelism that Module A measures, and it is not a two-locus problem. Crucially it is also *invisible* to any two-locus test: linkage disequilibrium between two loci is bounded by their allele frequencies, and as either locus approaches fixation the disequilibrium it can carry goes to zero. The more completely a locus sorts unidirectionally, the less LD it can express. Two-locus tests therefore cannot see the most strongly unidirectionally-sorted loci at all; those belong to Module A, and Module D is complementary rather than redundant.

(2) Bidirectional sorting (equal-fitness incompatibility). If the two parental combinations are *equally* fit while the recombinants are not, each replicate independently fixes one or the other compatible combination, chosen by drift. Per locus this looks like strong sorting in *random* directions — some populations toward *F. aquilonia*, some toward *F. polyctena* — i.e. bidirectional sorting. Its single-locus parallelism test is null by construction ($n_{\mathrm{aqu}} \approx n_{\mathrm{pol}}$), so Module A cannot flag it. The signal lives at the pair level, which is what the structure-corrected scan below is built to catch. Note that the among-population component of two-locus LD is *maximised* exactly here: a locus fixed in the same direction everywhere has no among-population variance, whereas one fixed in different directions across replicates has a great deal — so bidirectional loci are the natural home of the among-population DMI signal.

(3) Coupling maintained within populations. A third trace is elevated two-locus LD *within* populations (Ohta's $D^2_{IS} > D^2_{ST}$): particular allele combinations staying rarer than expected inside demes. This is the classic within-deme epistatic-selection signature, but it is badly confounded by population structure — in particular by the fact that our samples are workers from few colonies (nine of twenty "populations" are a single nest; nestmates are siblings), so cryptic relatedness inflates apparent within-population LD (a Wahlund effect). Any test of this trace must therefore control for relatedness.

The descriptive among-population screen reported separately established that the *raw* among-population LD between unlinked clusters is dominated by a single genome-wide ancestry gradient (drift/structure), with no pair specificity. The analysis below removes that confound with a mixed model that conditions on genome-wide relatedness (Section 2), addressing traces (2) and (3) at the individual level, but must first clear a methodological obstacle: cross-chromosome paralogy (Section 3).

2 Structure-corrected two-locus association (EMMAX)

Rationale and model. The cleanest way to ask whether two loci are associated *beyond* what shared ancestry and relatedness predict is a linear mixed model. We treat one LD cluster's consensus dosage as a quantitative trait and test its association against every other cluster,

$$y_i = \mu + \beta\, x_j + u + \varepsilon, \qquad u \sim N(0, \sigma_g^2 K), \quad \varepsilon \sim N(0, \sigma_e^2 I), \tag{1}$$

where y_i is the focal cluster's dosage across the 164 individuals, x_j a candidate partner, and K a genomic-relationship matrix (EMMAX; Li, Kemppainen, Rastas & Merilä, a method built for LD-clustered genome-wide data). This is a GWAS in which the "phenotype" is one locus and the "markers" are all the others. The variance components are estimated once under the null and reused for every partner, so a whole Manhattan is cheap.

Why one random effect handles both confounds. The leading eigenvectors of K are the ancestry/structure principal components, so conditioning on K removes the genome-wide ancestry gradient — the confound that swamped the descriptive screen — at the individual level. And because K also encodes the sibling relatedness among nestmates, the Wahlund inflation of within-population LD (trace 3 above) is regressed out in the *same* term. A residual peak between two loci is therefore pair-specific association with both the ancestry gradient and the colony structure already removed. Continuous eMLG dosages are used directly, with no hard-calling. Because the scan is symmetric we use *double* leave-one-chromosome-out: both the focal and the partner chromosome are dropped from K, so neither tested locus's own region deflates its association. This recovers a little power over single leave-one-out and leaves the test calibrated ($\lambda \approx 0.98$–1.05). One further point is worth stating, because it runs against intuition: K must be built from the ancestry-*informative* (differentiated) markers, not from a neutral, LD-pruned background. The confound K has to absorb here is the ancestry-sorting axis, which is a *selected* structure carried by those markers; a neutral basis (right for a demographic control such as the BayPass Ω) dilutes that eigenvector and under-corrects, inflating λ to $\sim$1.2 and leaking the ancestry gradient back as spurious positive associations. Finally, each focal trait is residualised on the top ten genome-wide PCs before the mixed model: K cannot remove a *secondary* structure axis that is concentrated on the very chromosomes double-LOCO drops, and such axes otherwise surface as hubs (Section 4). Conditioning on the PCs strips the dominant, climate-aligned axes — after it the random-focal control gives essentially no significant partner — while calibration holds. A first, targeted run scanned 51 focal loci (bidirectional candidates, extreme among-population-covariance loci, and controls); peaks were called at a Bonferroni genome-wide threshold, and each is signed (coupling vs repulsion) by its structure-corrected GLS coefficient — both cis and trans hits are recovered.

Reading a Manhattan. The self-peak at the focal locus and its local LD block are the built-in positive control that the pipeline works; the signal is any *additional* peak that is unlinked from the focal locus (different chromosome, or same chromosome $>$ 10 cM). The scan is well calibrated: genomic-inflation λ ranges 0.95–1.02 across focal sets and the background is flat (Fig. 1b, a control locus with only its self-peak). Fig. 1a shows a focal locus with a clean self-peak and a spread of unlinked peaks — exactly the structure-corrected pair-specific association we are after.

The first result is a warning. The top associations are correlated *within* populations (median within-population $r \approx 0.99$), robust to dropping the most influential individuals — so they are neither the ancestry axis nor a few-individual Wahlund artifact. But $r \approx 0.99$ between two unlinked loci, sustained inside every population, is not what an incompatibility makes; it is the signature of paralogy. This is treated next, and it is why the Manhattan in Fig. 1a distinguishes clean peaks (orange) from duplicate peaks (grey crosses): the focal locus's strongest partners are duplicates, its moderate partners are genuine candidates.

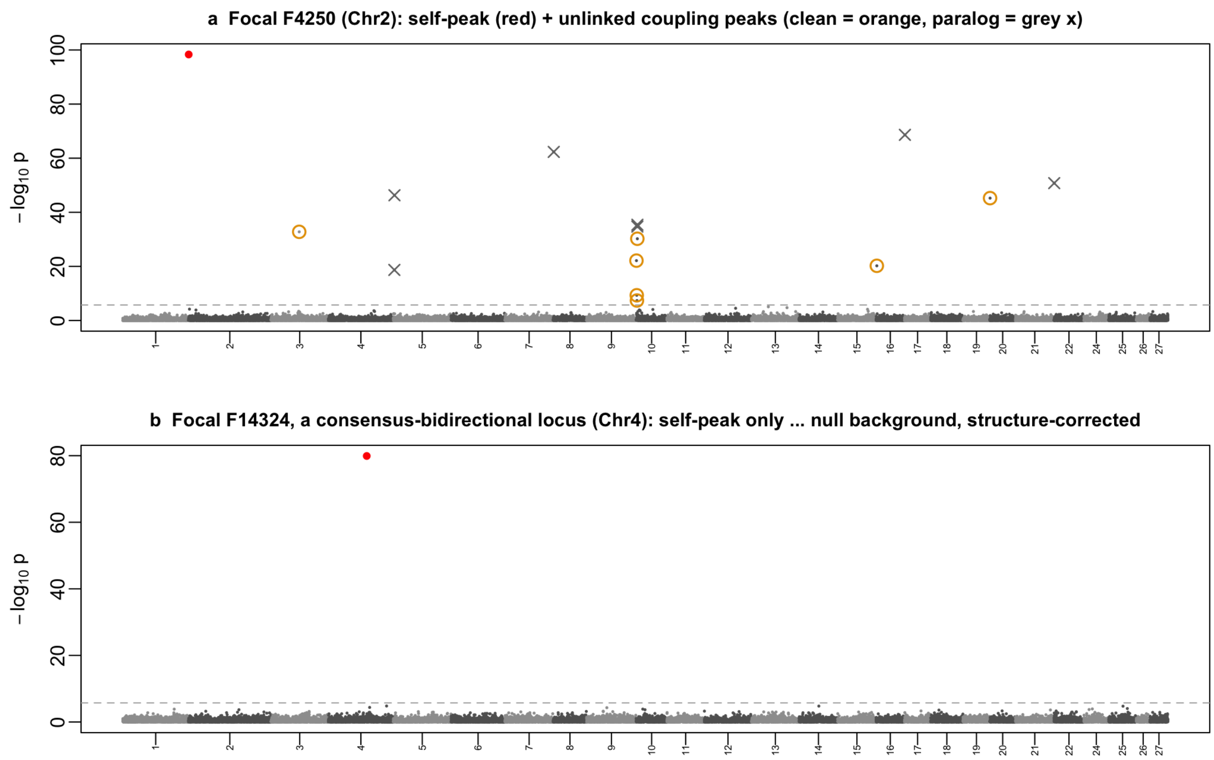

Figure 1: Structure-corrected two-locus association (EMMAX, double leave-one-chromosome-out K from the differentiated eMLGs). **(a)** A focal locus (F4250, Chr2) showing its self-peak (red) and unlinked partner peaks, split by the paralogy filter into genuine candidates (orange) and cross-chromosome duplicates (grey crosses). **(b)** A consensus-bidirectional focal locus (F14324, Chr4): only its self-peak, a flat structure-corrected background ($\lambda \approx 1$). Dashed line: Bonferroni threshold.

3 Cross-chromosome paralogy, and why distance cannot remove it

The problem. Two loci on different chromosomes are at infinite recombinational distance, so any strong LD between them cannot be linkage. It can be a genuine evolutionary correlation — but those are *moderate* — or it can be a technical artifact in which the two "loci" are not distinct at all. When a segmental duplication or a mis-assembly places one genomic region at two locations, reads from the single true region map to both, and the two "clusters" become genotype-identical by construction. Such pairs are the *loudest* inter-chromosomal disequilibrium in any dataset, and they are not biology. In the EMMAX run they reach $r^2 \approx 0.98$ and genotype concordance 1.000, and they recur on particular scaffolds (Chr26). A genetic-distance rule is powerless against them: different chromosomes already are maximal distance. (Distance does its job on the *within*-chromosome side — our conservative complexity-reduction merge at 0.5 cM rather leaves nearby clusters split than over-merges them, and the 10 cM unlinked threshold then excludes those split neighbours from long-range candidacy — but that is a separate problem.)

The discriminant. We flag duplicates by the median *within-population* correlation of the two clusters' dosages. Within-population removes the among-population (ancestry-axis) component, which is not what paralogy is about; what remains is individual-level identity. Genuine long-range coupling in these data sits at $|r| \lesssim 0.6$; a duplication sits at $|r| \approx 1$. A flipped or complementary duplicate (one copy allele-swapped relative to the other) gives $r \approx -1$, which the absolute value still catches. We flag pairs with median within-population $|r| > 0.9$. Two independent signals corroborate the flag: genotype *concordance* approaching 1 (positive duplicates), and per-cluster excess *heterozygosity* — a collapsed duplication makes reads from two paralogs look constitutively heterozygous, inflating H_o (Fig. 2b). The flag is deliberately simple so it can be re-thresholded from the stored values; and it is a screen, not a proof, because an extremely strong genuine within-deme incompatibility could in principle also reach $|r| \approx 1$ — which is why the corroborating signals matter.

It affects both arms, unequally. The filter is shared between the EMMAX and the among-population arms. It flags 32/332 EMMAX hits (10%) but only 55/401,945 among-population survivors (0.01%). The asymmetry is informative: a duplicate *is* structure-corrected individual-level LD, so it surfaces strongly in the mixed-model arm, whereas the among-population statistic — which asks about allele-frequency covariance across populations — is comparatively insensitive to it. In the among-population arm the excess-heterozygosity signal cleanly separates the flagged pairs ($H_o \approx 0.70$ vs 0.37 background); in the EMMAX arm the pre-selected focal set is already high-H_o, so within-population $|r|$ carries the discrimination and H_o is less informative (Fig. 2). Removing the flagged pairs leaves 300 clean EMMAX candidates. The filter is also a free by-product: it identifies duplicated/mis-assembled regions directly from the population data.

4 A systematic pass: global FDR, region-merging, and the candidate modules

The first, targeted run (Section 2) called peaks one focal Manhattan at a time under a per-focal Bonferroni threshold. To put every association on one footing — and to let the hubs and the pair-specific residue separate themselves under a single criterion — we re-read the whole EMMAX scan as one procedure. All 1.38×10^6 unlinked $(\text{focal}, \text{partner})$ tests were pooled and controlled at a global Benjamini–Hochberg $q < 0.01$ (which reproduces the earlier count, ~276 pairs, but as one whole-experiment error rate rather than 51 independent ones). We then (i) removed cross-locus paralogues ($|r_{\text{within}}| > 0.9$; −29 pairs); (ii) applied a *third-level merge* — for this network only, not the canonical clustering — grouping same-chromosome clusters within 10 cM by *average-linkage* clustering of their consensus correlation (cut at $|r| > 0.5$), so that one low-recombination region stops masquerading as a swarm of interacting loci. Average linkage requires the whole merged group to be mutually correlated; connected components (the naive choice) is single linkage and

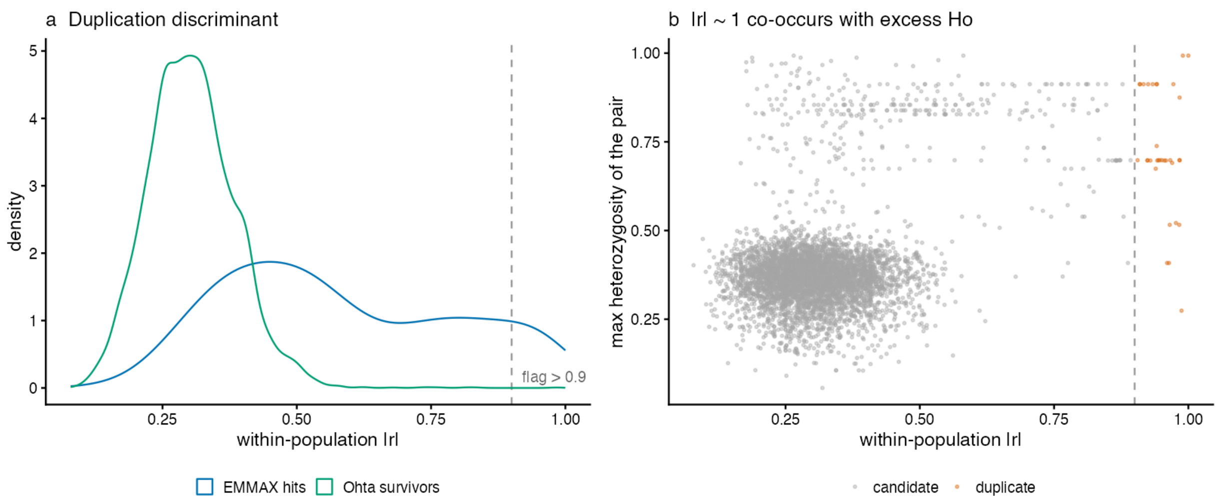

Figure 2: The paralogy discriminant. **(a)** Distribution of within-population $|r|$ for candidate pairs: the among-population survivors (green) concentrate at moderate $|r|$ with a thin tail, whereas the EMMAX hits (blue) carry a heavy tail to $|r| \approx 1$ — the duplicates. Dashed line: the 0.9 flag. **(b)** Pairs at $|r| \approx 1$ (orange, flagged) also carry elevated heterozygosity, corroborating a duplication origin; excess H_o alone is not sufficient (many high-H_o pairs at moderate $|r|$ are genuine).

chains uncorrelated ends together, so a 16-cluster block on Chr26 that single linkage swallows whole instead resolves into a few mutually-correlated sub-blocks (156 $\rightarrow$ 127 meta-nodes). We then (iii) dropped edges whose among-population allele-frequency correlation does not survive leave-one-population-out ($|r|_{\mathrm{LOO}} < 0.3$; -50 edges), which by itself dissolved a spurious degree-19 hub on Chr4 built from a single deme — exactly the near-fixed / leverage artifact the MAF–LD bound predicts. Low-recombination meta-nodes (recombination percentile < 0.1) were *labelled* "structure" rather than removed. What remains is 110 meta-edges over 83 meta-nodes (18 trans, 92 cis; Fig. 4).

The network splits cleanly into two kinds of object. The dominant one is a single grey *structure* complex — the low-recombination hubs (the merged F33028/F33095 anchor and F27700, plus F57361) wired almost entirely to one another: 66 of the 110 edges are structure–structure. A PCA over the 87 loci of this module (Fig. 7) shows what it is *not*. These loci separate individuals by population *less* than a random draw of differentiated loci (mean per-locus among-population $R^2 = 0.43$, against 0.54 for random differentiated and 0.64 for the candidate loci), and they load only weakly on the genome-wide PCs ($|r| \leq 0.23$). Their cross-genome correlation is therefore neither among-population geographic differentiation nor the dominant ancestry axis; it is *individual-level admixture-block covariance* — ancestral haplotype blocks co-inherited within individuals and, in low-recombination regions, not yet broken up — carried within populations. These loci also sit in the lowest recombination percentile (centromeric and pericentromeric), which is why they persist through PC-conditioning.

Low recombination is a caveat, not a verdict. We label these regions "structure" as an *annotation*, never a filter, because low recombination does not imply neutrality — it is exactly where coadapted/epistatic complexes are expected to accumulate and persist (recombination suppression is a hallmark of coadaptation). For an unlinked pair the between-region recombination fraction is 0.5

regardless of local rate, so low local recombination does not "hold" cross-chromosome LD; it makes each region a clean, low-noise ancestry *block*, i.e. the highest-signal marker of whatever residual covariation exists after PC-conditioning — finer neutral structure *or* selection. The PCA above is *consistent with* neutral relic co-ancestry but does not exclude selection (parallel or stabilising selection also lowers differentiation). The only clean discriminant is a recombination-matched neutral null (Module E, Section 5): every locus, low-recombination included, is carried into that null and judged against its own local-rate baseline, so a low-recombination pair whose LD *exceeds* even that high baseline earns its way back as a coadaptation candidate.

That the structure "module" is not one block but several is confirmed directly. An average-linkage clustering of the meta-node consensuses *across* chromosomes (cut at $|r| > 0.4$; distinct from the within-chromosome merge, which only collapses redundant neighbours) resolves the network into 23 co-ancestry modules, of which a handful are substantial: a dominant module of 16 meta-nodes spanning 12 chromosomes, and secondary modules of 14, 9 and 7 meta-nodes. Plotted one module at a time (Fig. 3), each is a single coherent block — across all its SNPs on six or more chromosomes the individuals take one of the three dosages *together*, exactly as a co-inherited admixture haplotype block should, and unrelated to the other modules the all-structure view had mixed in. Drawing only the within-module links on the circos (79 of the 110 edges) makes the same point genome-wide: the co-ancestry resolves into a few disjoint colour-coded bundles, not one hairball. The candidate F33454 aquilonia module below falls out as its own module here too.

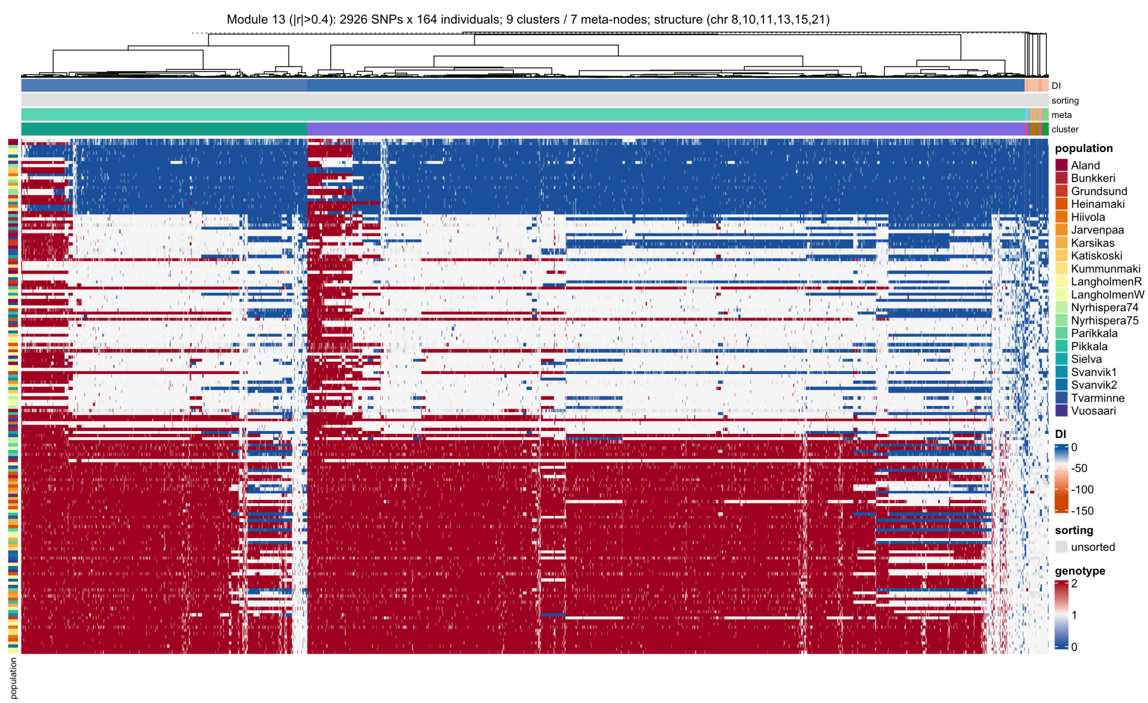

Figure 3: One cross-chromosome co-ancestry module (9 clusters / 7 meta-nodes across Chr8/10/11/13/15/21), as a genotype heatmap of its member SNPs (columns ward.D2-clustered; colour bars: cluster, meta-node, sorting, DI; rows = individuals along the module's co-ancestry PC1). The module behaves as a *single* co-inherited haplotype block: an individual is homozygous-reference (blue), heterozygous (pale), or homozygous-alternate (red) consistently across all 2926 SNPs on all six chromosomes, and the three groups do not coincide with populations — individual-level admixture-block co-ancestry, not geographic structure or a pairwise incompatibility.

The gold *candidate* nodes are the residue of interest, and they are not merely isolated pairs. Thirteen edges have both endpoints outside the structure module (Table 1), and they organise into three kinds: (a) an *aquilonia co-sorting module* anchored on F33454 (Chr10, 50 cM, degree 12), whose partners on Chr3/4/5/9/11/13/19 are almost all aquilonia-sorted — coordinated ancestry co-selection toward the minor (aquilonia) background rather than an obvious pairwise incompatibility; (b) a single strong, ancestry-*independent* trans pair, F11431–F49480 (Chr3–Chr16, $q = 2.2 \times 10^{-10}$, both loci unsorted) — the cleanest pairwise-DMI candidate the network contains; and (c) a handful of same-ancestry *cis* pairs (polyctena: F17114–F49429, F15820–F26284; aquilonia: F22294–F39587), i.e. co-sorting, not repulsion. Stripped of structure and of ancestry co-sorting, the pair-specific residue is thus F11431–F49480 together with a couple of weak, leverage-stable trans pairs — a handful of edges in all, reached by one reproducible criterion.

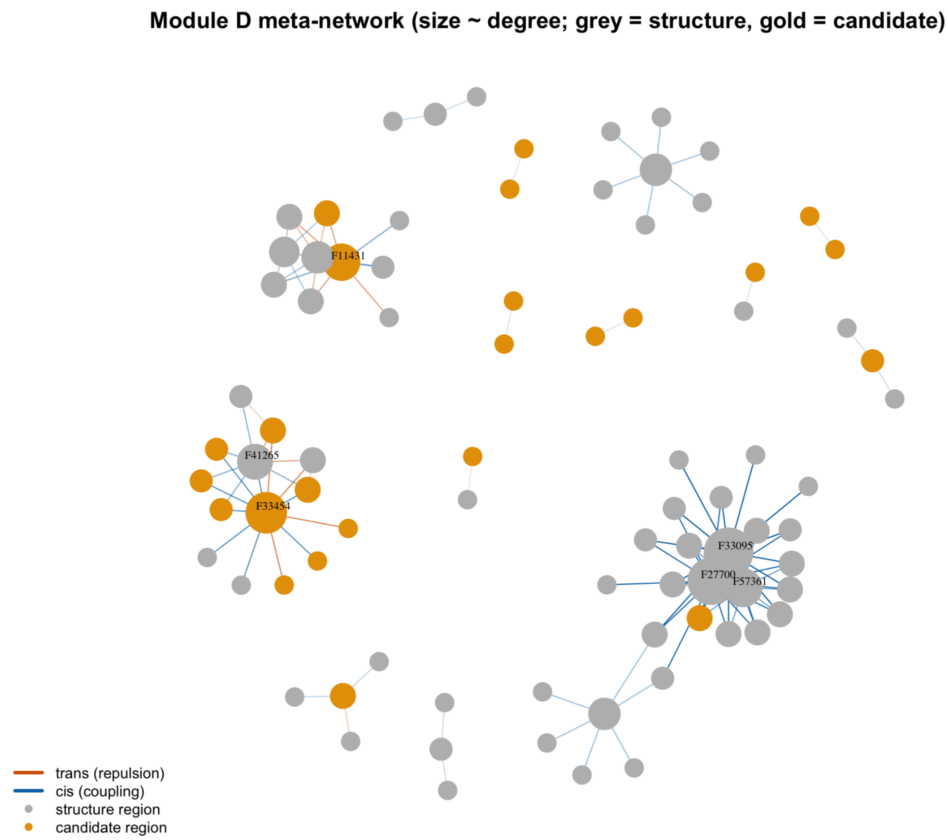

Figure 4: The Module-D meta-network after global FDR ($q < 0.01$), paralogy removal, third-level within-region merging, and the leave-one-population-out leverage filter. Nodes are meta-nodes sized by degree; grey = low-recombination "structure" label, gold = candidate region; blue edges cis (coupling), orange edges trans (repulsion). The large grey complex (F33095/F27700/F57361) is the admixture-block structure module; the gold hub F33454 is the aquilonia co-sorting module; scattered gold–gold pairs are the pair-specific candidates.

Table 1: The 13 candidate–candidate meta-edges (both endpoints outside the structure module), with chromosomes, coupling, FDR q, and the ancestry-sorting class of each endpoint. The F33454 anchor and its aquilonia-sorted partners form a co-sorting module; F11431–F49480 is the one strong ancestry-independent trans pair.

Pair	Chr	coupling	q	sorting (a/b)
F11431–F49480	3/16	trans	2.2×10^{-10}	unsorted/unsorted
F10974–F33454	3/10	trans	2.7×10^{-4}	aqu/aqu
F32802–F33454	9/10	trans	1.1×10^{-3}	aqu/aqu
F11265–F33454	3/10	trans	6.2×10^{-3}	pol/aqu
F15419–F33454	4/10	cis	2.8×10^{-11}	aqu/aqu
F33454–F57178	10/19	cis	8.8×10^{-5}	aqu/aqu
F33454–F41301	10/13	cis	1.2×10^{-4}	aqu/aqu
F33454–F35575	10/11	cis	2.2×10^{-4}	aqu/aqu
F17114–F49429	4/15	cis	9.2×10^{-4}	pol/pol
F15820–F26284	4/7	cis	2.3×10^{-3}	pol/pol
F21099–F33454	5/10	cis	3.0×10^{-3}	aqu/aqu
F22294–F39587	6/12	cis	4.7×10^{-3}	aqu/aqu
F45183–F45504	14/14	cis	5.0×10^{-3}	unsorted/unsorted

5 Synthesis and what the simulation will settle

One structure-corrected scan, read out by a single reproducible procedure (global FDR, average-linkage region-merging, and a leave-one-population-out leverage filter), gives a largely negative result. Paralogy, the dominant ancestry axis, single-deme leverage, and near-fixed loci are each removed in turn, and every removal exposes an apparent signal as an artifact rather than an incompatibility. Two kinds of object remain. The bulk is a low-recombination *structure* module — unlinked regions co-inherited as individual-level admixture blocks, which the genotype PCA shows separate populations *less* than random differentiated loci and load only weakly on the genome PCs. The residue of interest is a short candidate list: an aquilonia *co-sorting* module (F33454 and partners) that belongs with the ancestry-sorting arms A/C, a single strong ancestry-independent trans pair (F11431–F49480), and a couple of weak, leverage-stable trans pairs.

This is the expected place to be *before* the null. Every control here removes structure *aligned* with what it conditions on; none can say whether the residue — or, crucially, any of the low-recombination modules — exceeds what neutral, recombination-matched drift would produce in twenty replicate populations of this size and sampling design. That is Module E, and the revision that matters most is how its null is built.

Low recombination must be judged, not excluded. Low-recombination regions are exactly where coadapted complexes are expected to accumulate, so the "structure" label is an annotation, not a filter: *every* locus is carried into the null. The null itself must be matched to *local* recombination — neutral admixture simulated on the real recombination map, with the same founder ancestries, generations and sampling — so that a low-recombination pair is compared against its own, legitimately high, neutral baseline and can still surface if its LD exceeds it. Two corroborations are available even before the simulation, and both separate selection from neutral relic co-ancestry: *replicate-population parallelism* (neutral relic LD drifts in random directions across the twenty populations, whereas epistatic selection maintains the same combination reproducibly) and *specific exceedance* of a module over the general low-recombination background. The coherent multi-chromosome blocks — where an individual takes one dosage across all of a module's SNPs on six or more chromosomes (Fig. 3) — are the first coadaptation candidates to test, not the ones to set aside. Module E converts each of these leads into an excess-over-neutral statement, or not; until then the result is descriptive.

Appendix: explored and set aside

For the record, several analyses were run and are *not* part of the minimal pipeline (see `moduleD_plan.md`). The Ohta among-population LD arm as a standalone screen: the raw among-population signal is the ancestry cline, so it is retained only as Module E's measurement instrument. An SNP-level bidirectional-sorting test: a weak, diffuse excess of co-sorting whose per-locus permutation null overstates significance by not modelling shared drift. A per-focal Bonferroni read-out: replaced by one global FDR. A neutral/LD-pruned GRM basis: it under-corrects, because K must *contain* the selected ancestry axis it removes. And a climate-association tie for the hub loci: negative — they are among the least climate-associated loci in the genome, so the hubs belong to neither the intrinsic nor the extrinsic arm. These are recorded so the pruning is deliberate, not lost.

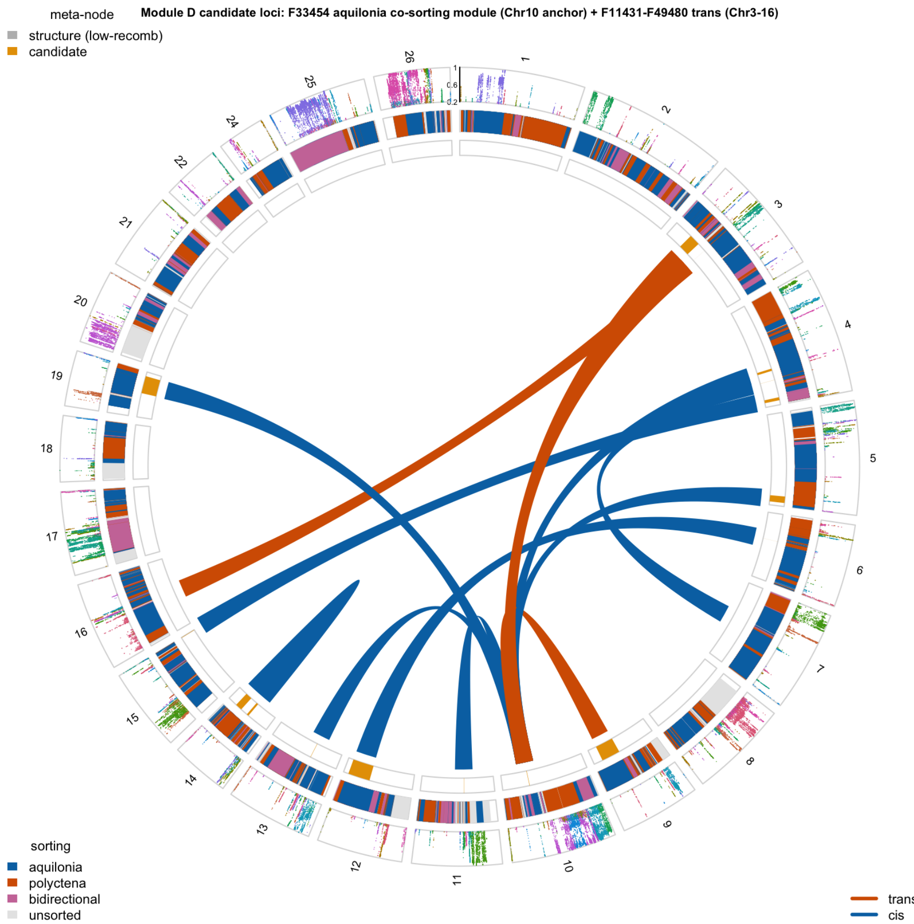

Figure 5: Genomic view of the candidate loci: each association drawn as a band spanning the full extent of the two meta-nodes it joins (cis blue, trans orange; here at full opacity as the set is sparse, with a minimum band width enforced so narrow regions stay visible). The Chr10 anchor F33454 fans out to aquilonia-sorted partners on Chr3/4/5/9/11/13/19 (bands landing on the blue aquilonia stretches of the sorting ring) – the aquilonia co-sorting module – while the single orange band F11431–F49480 (Chr3–Chr16) is the one ancestry-independent trans pair. Outer track: $\mathrm{ld_w} > 0.2$ per marker; inner rings: genome-wide sorting, then the meta-node structure(grey)/candidate(gold) label.

Figure 6: Genomic view of the structure / co-ancestry module (band opacity $\propto$ degree; black = the F27700–F33095 heatmap pair). The module is almost entirely *cis* (blue): low-recombination admixture blocks on different chromosomes couple in the *same* ancestry direction, converging on the Chr8/Chr10 anchors. The contrast with Fig. 5 – cis-coupled co-ancestry here, the trans/repulsion signal with the candidates – is the visual statement that the two are different phenomena.

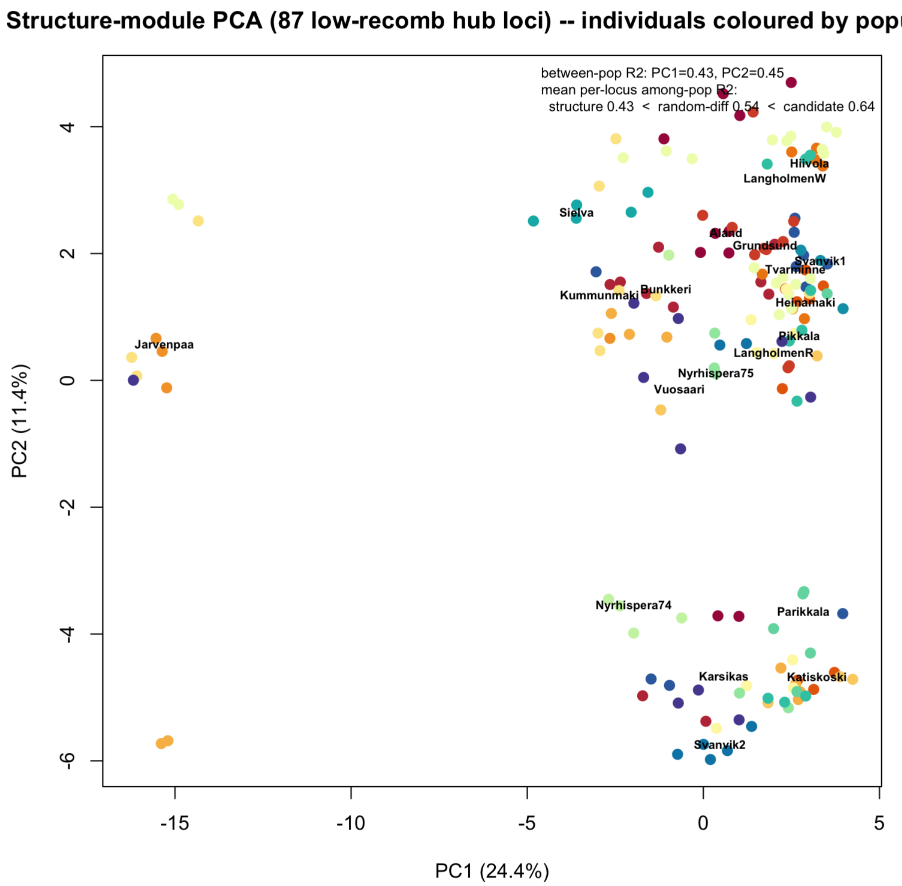

Figure 7: PCA over the 87 loci of the structure module, individuals coloured by population. The module carries real between-population structure (some populations, e.g. Järvenpää, pull out as coherent clusters), but *less* than a random draw of differentiated loci (mean per-locus among-population $R^2 = 0.43$ vs 0.54), and its axes correlate only weakly with the genome-wide PCs. The hub LD is individual-level admixture-block covariance in low-recombination regions, not geographic differentiation and not a pairwise incompatibility.